AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 25 · STUDENT LESSON

Text as Pieces

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Read the vocabulary. Match exactly. Count under a declared convention.

Trace chunk/character encodings, preserve source text and report unsupported input honestly.

Keep one habit: Vocabulary and IDs → longest-match trace → exact/partial decoding → changed vocabulary comparison.

MAIN LESSON · SECTION 1

What counts as a token under this exact vocabulary?

> Name the matching rule and preserve every supported source character.

Your learning goals

I can encode and decode text under an explicit toy **vocabulary**.

I can explain how segmentation changes sequence length.

I can handle unsupported characters without silently deleting information.

Recall exact source preservation

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Why must an exact representation preserve the original spelling, spaces and punctuation rather than silently changing them?

Your end goal

By the end, I can encode and decode text using a specified **vocabulary** and matching rule.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

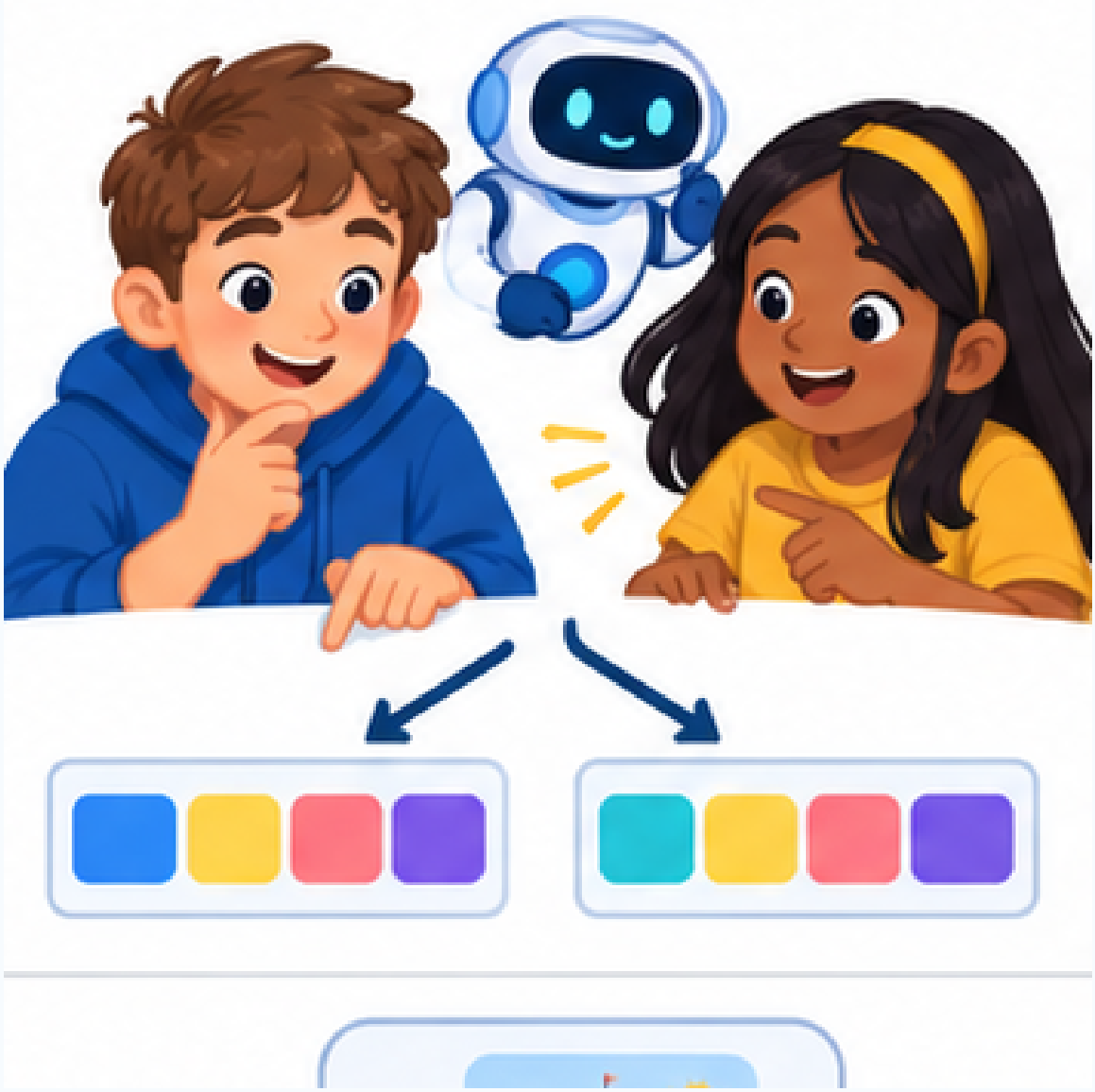
Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 25: Text as Pieces

Language systems **process token**s and **pattern**s rather than whole human meanings.



THINKER MISSION

Tokenize tricky phrases and compare alternate splits.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Read the case-sensitive vocabulary and longest-match contract

> Token IDs name entries; they are not meanings or similarity scores.

The complete toy tokenizer rule

Tokens need not be words. A classroom **token**izer can use an exact, case-sensitive **vocabulary** and greedily choose the longest matching **token** from left to right. This is a toy rule, not a claim about every production **token**izer.

Define IDs: 0=end, 1=cat, 2=c, 3=a, 4=t, 5=s, 6=space, 7=sat. The space **token** represents one literal space. The end **token** is a special terminator, not ordinary text.

At each position choose the longest matching ordinary **token**. If no **token** matches, stop and report the unsupported character and its position. Do not delete the character or guess a replacement.

IDs identify entries; a larger ID does not mean a **token** is more valuable or more similar.

The ID and special-token reference

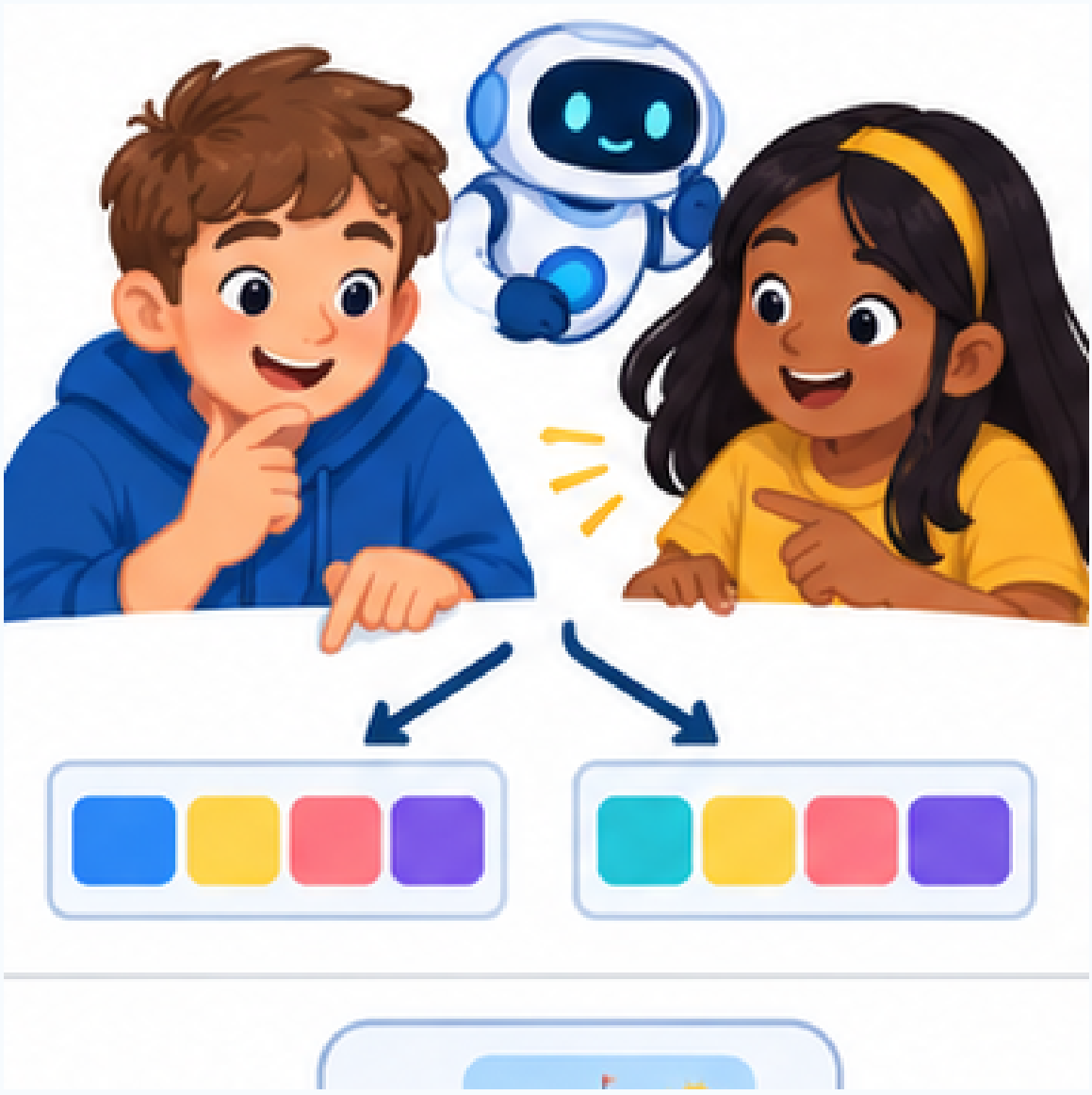
ID	Entry	Meaning
0	end	Special terminator; not an ordinary text match
1	cat	Three ordinary characters
2 / 3 / 4 / 5	c / a / t / s	Four separate single-character entries
6	one literal space	Preserve the actual space character
7	sat	Three ordinary characters

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use text as pieces to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

MAIN LESSON · SECTION 3

Trace encoding and exact decoding before counting

> The end marker changes sequence length only when the protocol adds it.

Follow the four longest matches

Remaining text	Chosen token	ID
cats sat	cat	1
s sat	s	5
sat	space	6
sat	sat	7

Decode and state the counting convention

The **encoding** of cats sat is [1,5,6,7], followed by [0] only if the protocol appends an end **token**. **Decoding** concatenates the ordinary **token** strings to recover cats sat.

Count four text **token**s, or five sequence items with the end marker included; always state the counting convention. The two written words do not imply two **token**s.

Compare a supported round trip with uppercase failure

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Exact text: cats sat	Longest-match encoding	IDs 1, 5, 6, 7
IDs 1, 5, 6, 7	Concatenate token strings	Exact text: cats sat
Exact text: Cat	No match for uppercase C	Report unsupported input

Encoding preserves supported characters and makes unsupported input explicit.

Explain a pictured step

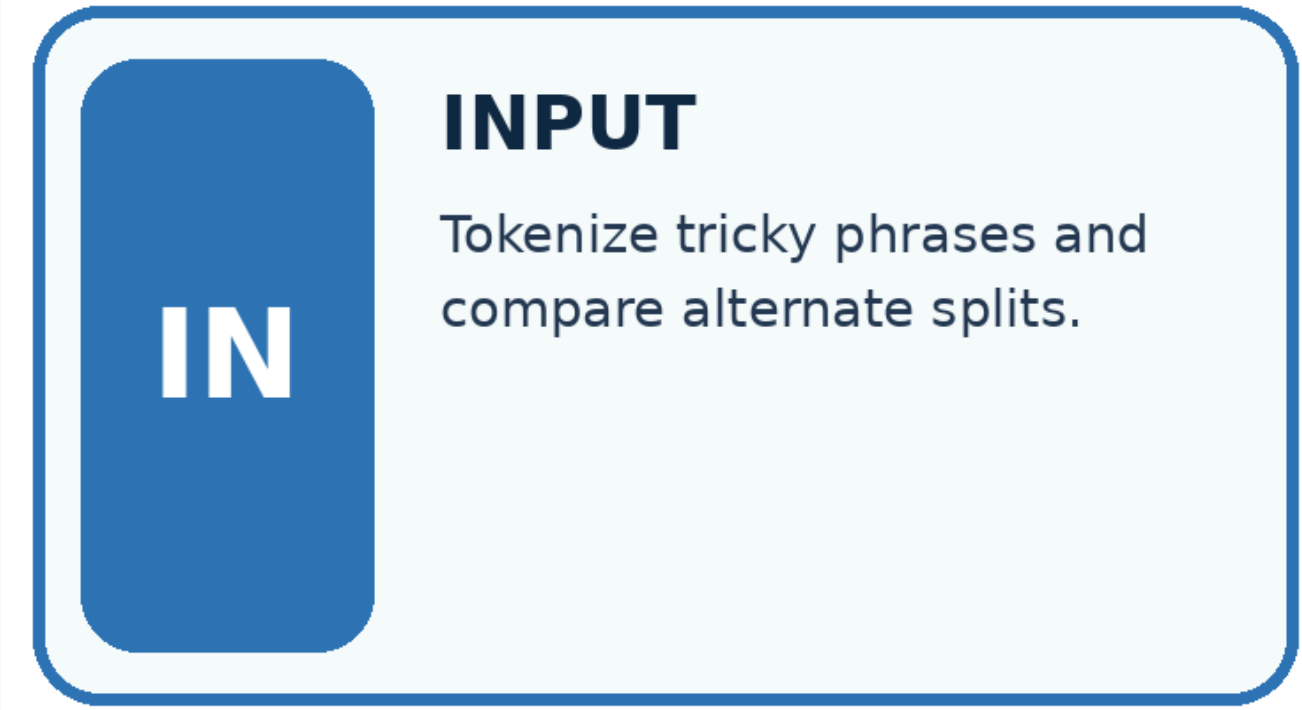
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
Language systems process tokens and patterns rather than whole human meanings.
OUT
OUTPUT
Tokenize tricky phrases and compare alternate splits.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

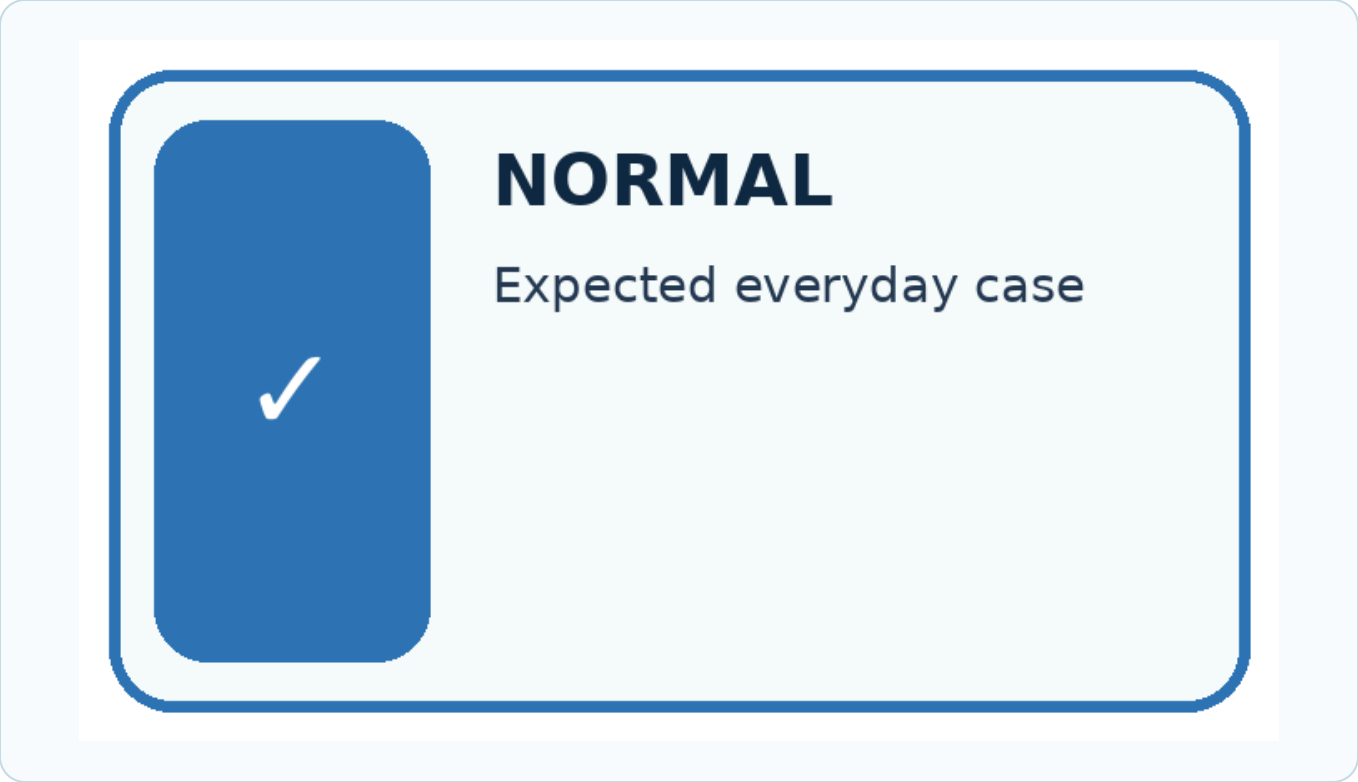
Connect the picture and the explanation

MAIN LESSON · SECTION 4

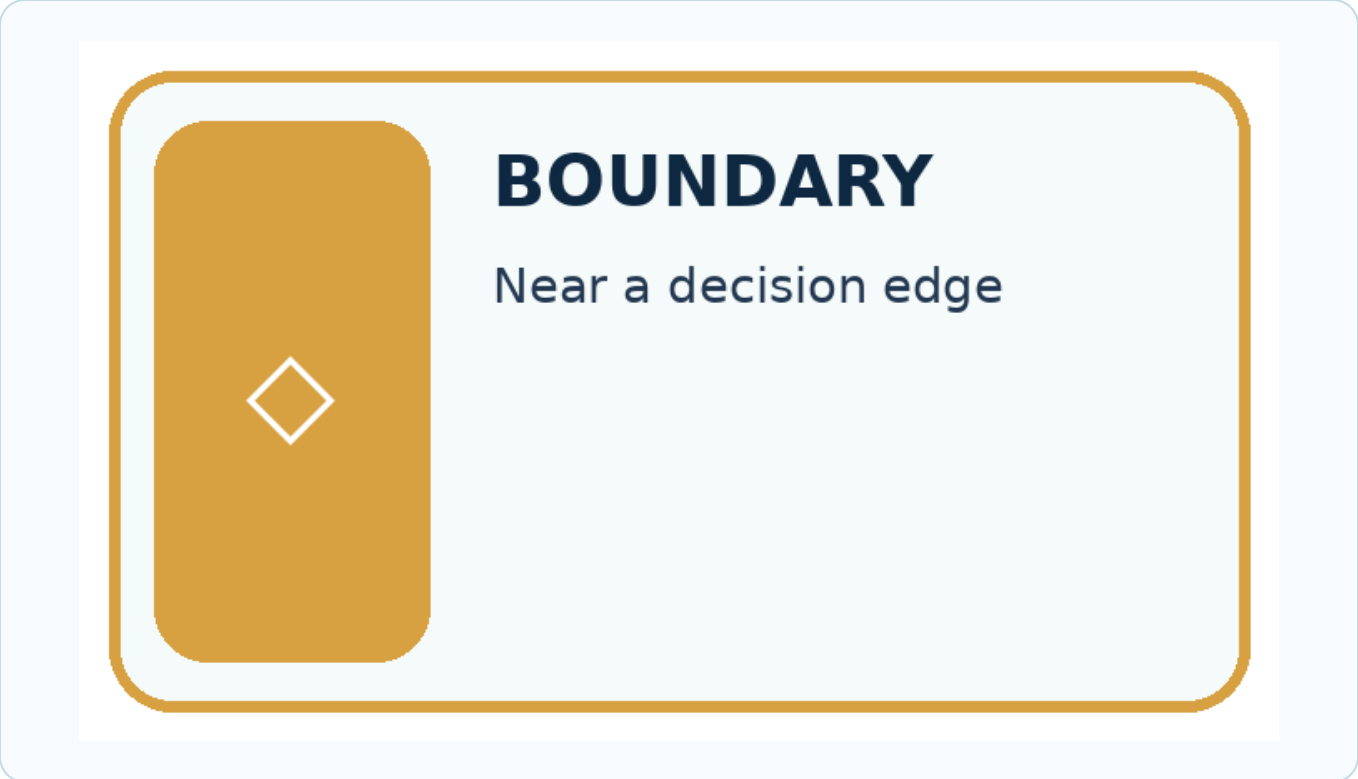
Check spaces, case and unknown characters

> A partial trace is not complete encoding success.

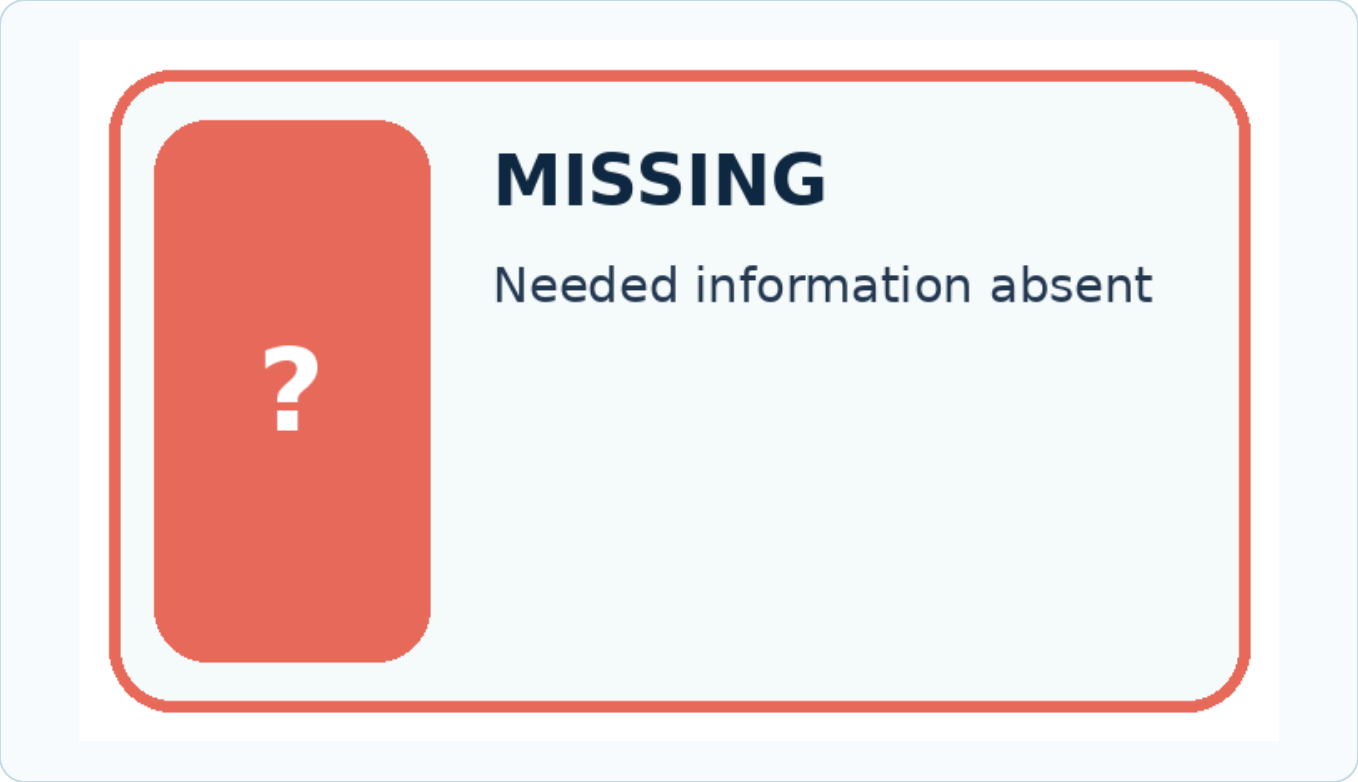
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Use a character **vocabulary** covering b,o,k,s and space instead.

▼ Check after predicting

Prediction: The same phrase uses ten character **token**s.

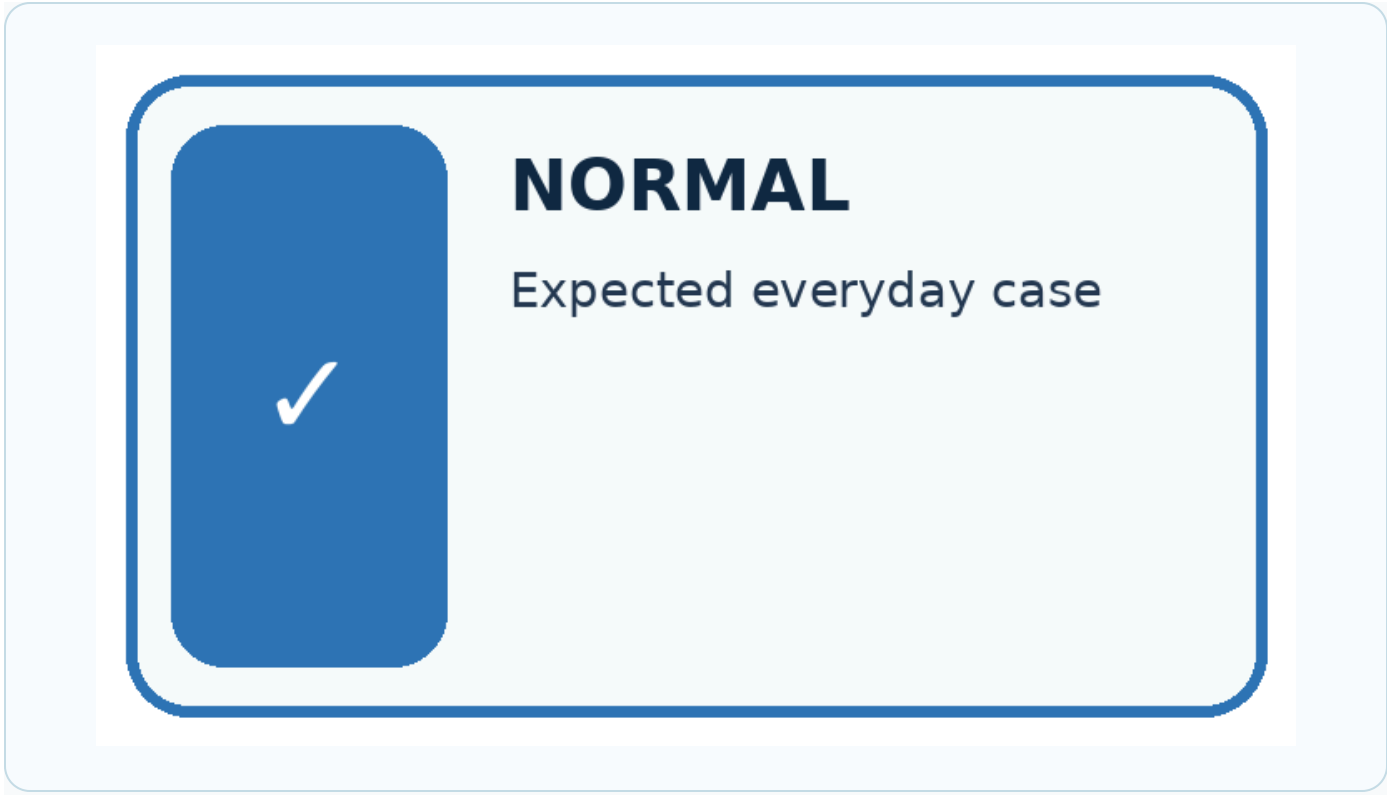
Reason: There are five letters, one space and four letters; its decoded text remains unchanged.

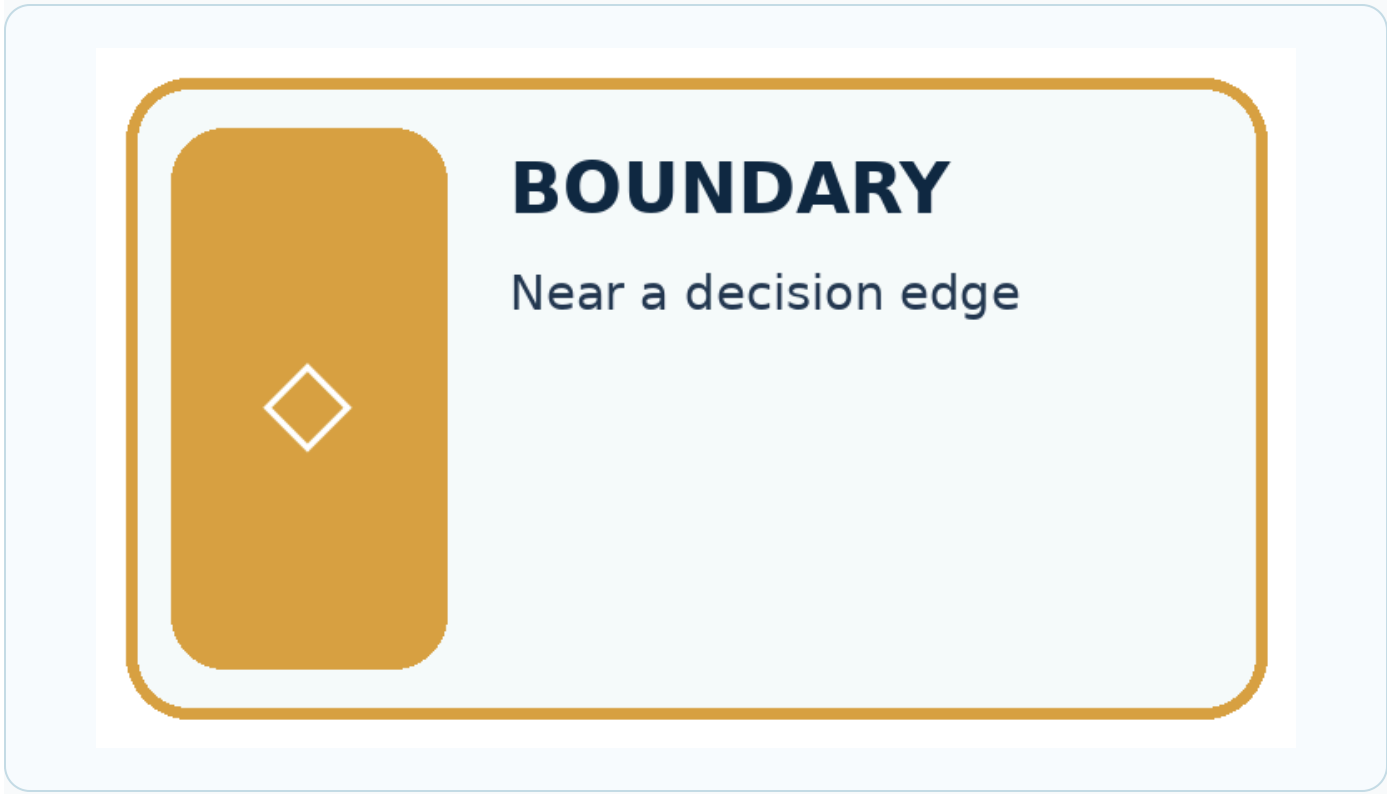
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

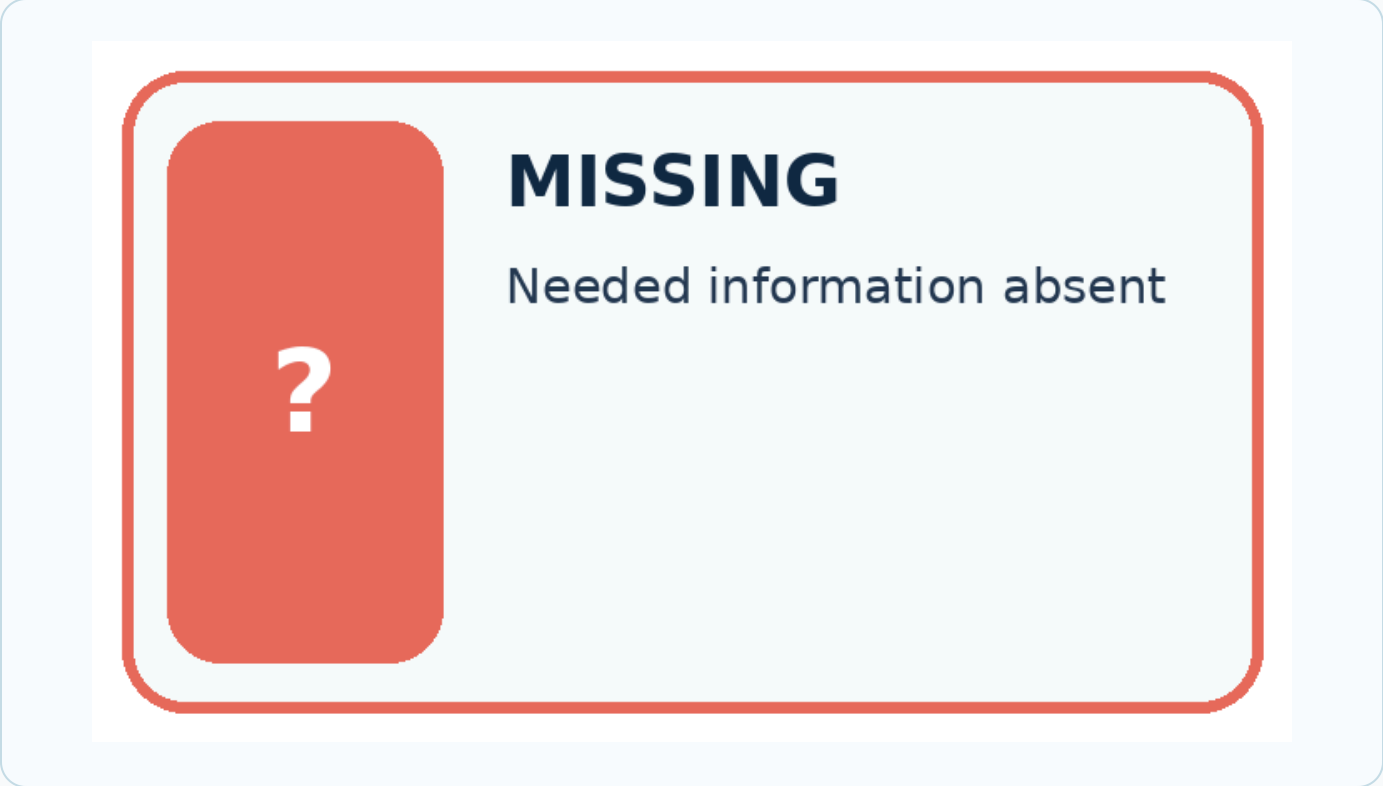
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Compare vocabularies, then a separate book phrase

> Do not mix the cat vocabulary with the book/books vocabulary.

Count character-only versus original cat tokens

A character-only **vocabulary** c,a,t,s,space could encode cats sat as c,a,t,s,space,s,a,t: eight text **token**s. The original **vocabulary** uses four. Both decode to the same source text.

Sequence length is a property of the text, **vocabulary** and **encoding** rule together. Counts from different **token**izers are not automatically comparable.

In this **vocabulary**, sat is chosen over s because it is longer. The toy longest-match rule is defined explicitly. Real **token**izers may use different algorithms, byte **encoding**s and special-**token** policies.

Never infer an unknown **token**izer's count from word count alone.

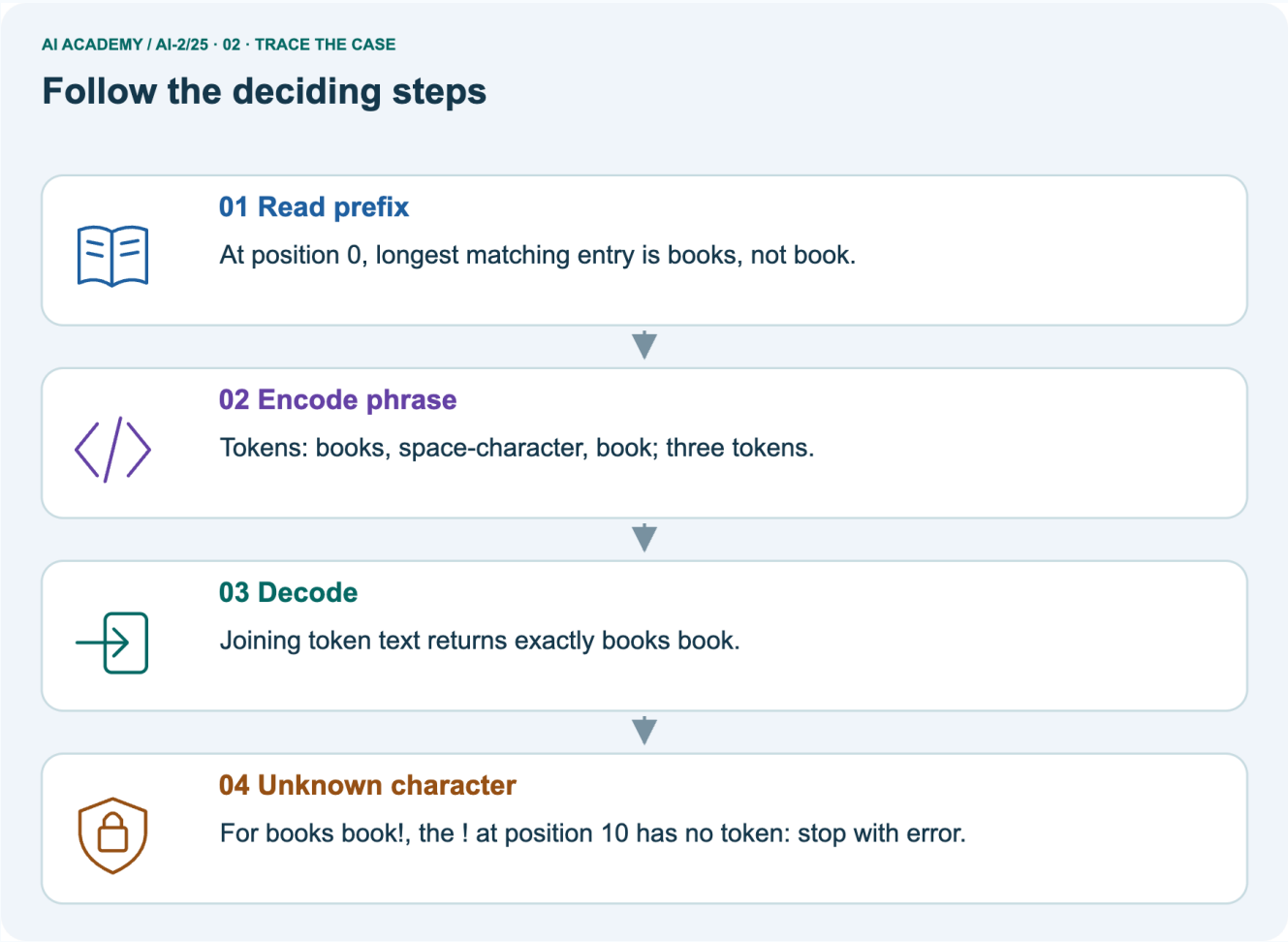
Read the separate phrase and its vocabulary

A fictional **token**izer converts a short catalogue phrase into **token**s.

Vocabulary entries: book, books and the space character. At each position choose the longest matching entry; preserve the actual space.

Text is "books book". Exclamation marks have no **token**; zero-based character positions are used.

Trace overlap, space and unsupported punctuation



Follow the labelled picture with the matching explanation. Open the vector diagram

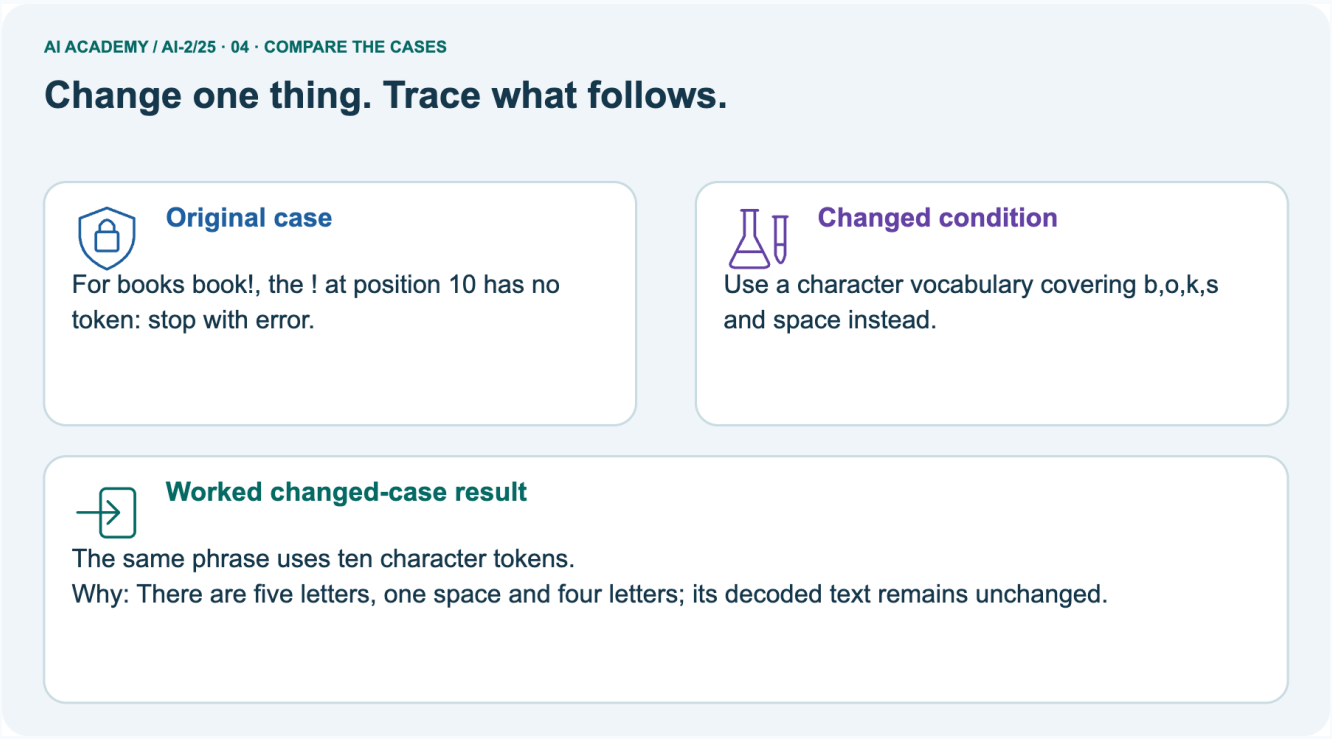
A **token** is defined by the **vocabulary** and **encoding** rule; it need not be one word or one character. Longest matching matters where **vocabulary** entries overlap. Counting text **token**s here does not count letters.

Preserve spaces during **decoding** so the reconstructed text matches the original exactly.

Predict the character-only book representation

- Change one thing: Use a character **vocabulary** covering b,o,k,s and space instead. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the zero-based failure position and limits

- Read the labelled picture: Why is the unknown ! at position 10 rather than position 9?
- This deterministic toy **token**izer omits real subword **training**, Unicode handling and model-specific special **token**s.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 25 • WORKED EVIDENCE

1. ‘unhelpful’ might split as ‘un’ + ‘help’ + ‘ful’.
2. A name or emoji may use unexpected token pieces.
3. Token boundaries need not match whole words.
4. Meaning emerges from patterns across context, not from one token acting like a dictionary.

1. ‘unhelpful’ might split as ‘un’ + ‘help’ + ‘ful’.

2. A name or emoji may use unexpected **token** pieces.

3. **Token** boundaries need not match whole words.

4. Meaning emerges from **pattern**s across **context**, not from one **token** acting like a dictionary.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

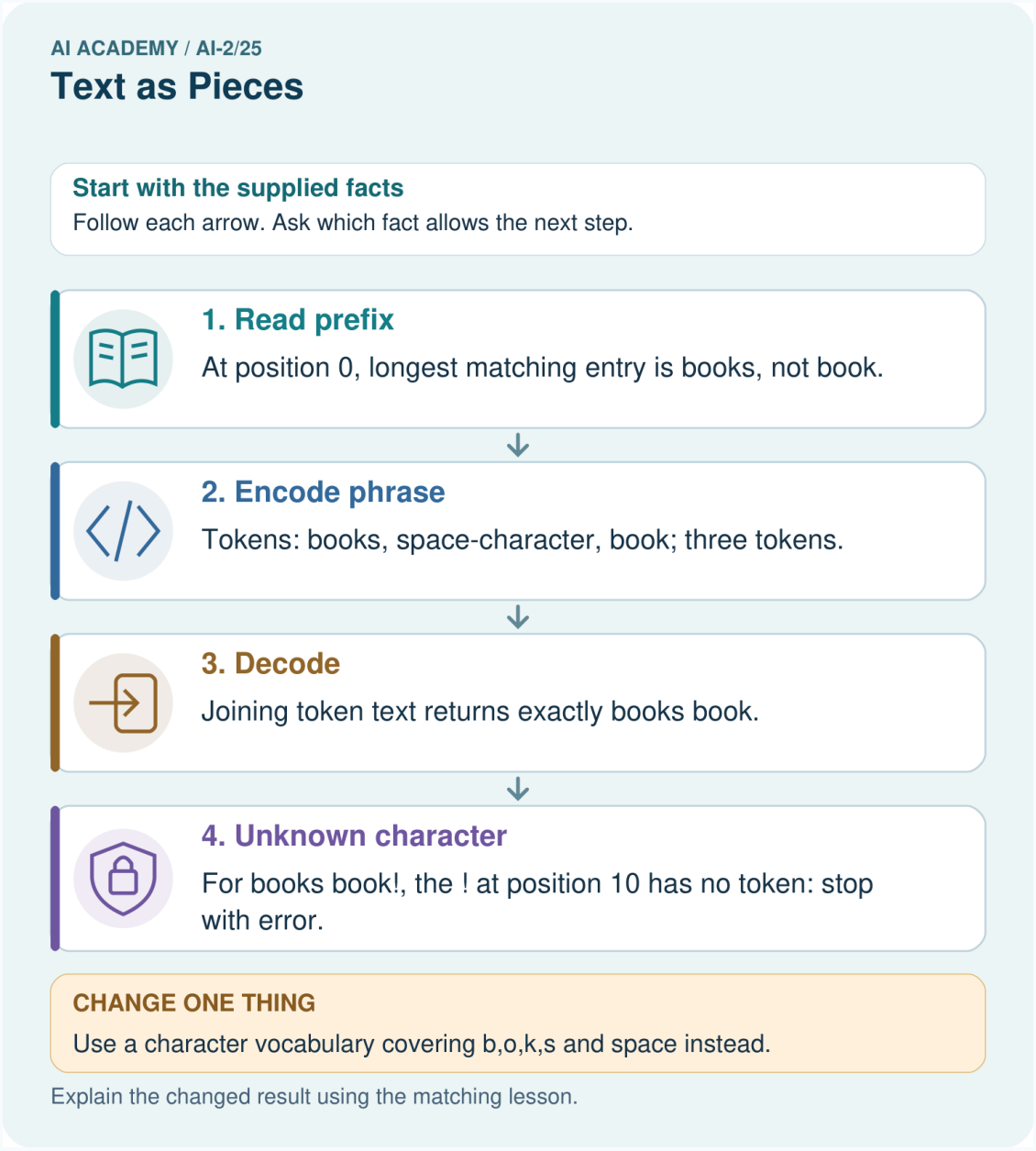
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Text as Pieces

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read prefix	At position 0, longest matching entry is books, not book.
2. Encode phrase	Tokens: books, space-character, book; three tokens.
3. Decode	Joining token text returns exactly books book.
4. Unknown character	For books book!, the ! at position 10 has no token: stop with error.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

A correct round trip verifies representation, not factual truth

> A silent source change is not an exact encoding.

Keep vocabulary, preprocessing and counting conventions

Decoding round trips check representation, not factual meaning. A supported spelling can be false; an unsupported character can be important.

Record the **vocabulary** version, rule and special-**token** convention when comparing counts or reproducing model **input**s. Whitespace, case and punctuation are part of this exact toy **input** contract.

Challenge dropping unsupported punctuation

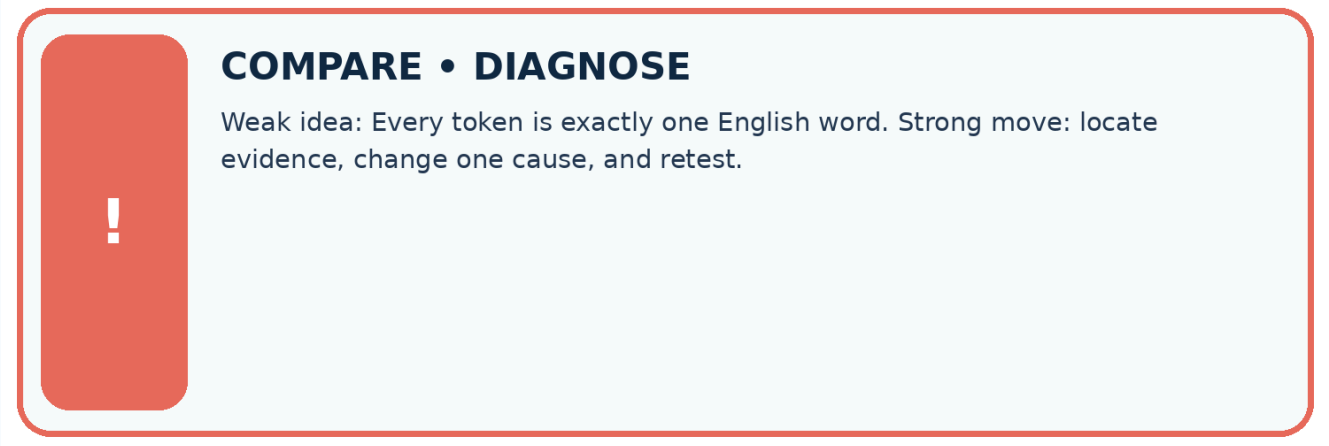
> **Exit: a program drops ! from cat! and announces a successful round trip. Identify the changed information and the honest response.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.



Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 8

Trace spaces and failure positions explicitly

> Keep the partial IDs and the reason when encoding stops.

Read the supported and failing examples

Supported case: cat encodes [1]; c at encodes [2,6,3,4]. A space changes both the character sequence and the possible matches. The **input** Cat is unsupported at position 0 because uppercase C has no entry.

The **input** cat! encodes cat, then fails on ! at position 3; a partial **encoding** must not be reported as complete success.

Try the guided c at trace

Guided: encode c at, count its text tokens and explain how cat! should be reported.

Encode and decode the W2 phrase

Use workbook W2 for the connected example "**Token**isation and **vocabulary**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair token always equals word

- Use the worked case for **Token**isation and **vocabulary**. Proposed explanation: "A **token** always equals a word". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the separate book failure index

Why is the unknown ! at position 10 rather than position 9?

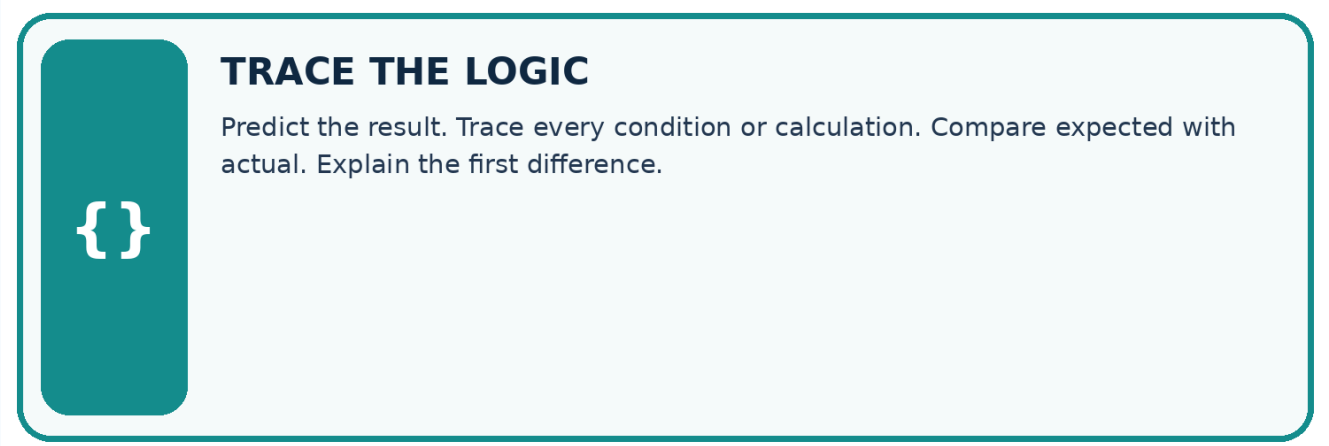
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Encode and decode

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Audit exact encodings, failures and a changed vocabulary

> Compare representations without silently altering the source.

The complete toy encoder and all investigation inputs

```
vocabulary = {1: "cat", 2: "c", 3: "a", 4: "t", 5: "s", 6: " ", 7: "sat"}
def encode(text, entries):
    ids = []
    position = 0
    order = sorted(entries, key=lambda number: (-len(entries[number]), number))
    while position < len(text):
        found = None
        for token_id in order:
            if text.startswith(entries[token_id], position):
                found = token_id
                break
        if found is None:
            return ids, (position, text[position])
        ids.append(found)
        position += len(entries[found])
    return ids, None
```

```
for text in ["cats sat", "c at", "Cat", "cat!"]:
    ids, error = encode(text, vocabulary)
    print(repr(text), ids, error)
    if error is None:
        decoded = "".join(vocabulary[token_id] for token_id in ids)
        print("decoded", repr(decoded), "text tokens", len(ids))
character_entries = {number: text for number, text in vocabulary.items() if
len(text) == 1}
ids, error = encode("cats sat", character_entries)
print("character-only", ids, len(ids), error)
```

Predict, trace, execute and compare

1. Predict every ID, preserved space and error position for the four supplied strings before execution. Use the complete program or label a paper trace. entries must contain unique positive IDs with nonempty ordinary token strings; special end ID0 is deliberately outside this matching table. Longest entries are tried first, then ID for an otherwise equal-length order. startswith checks an exact prefix at position; on success position moves by token length, and failure returns the partial IDs plus zero-based position/character. The decoder shown accepts only successfully encoded ordinary IDs, not an appended terminator; end handling is a separate sequence protocol. Compare the original and character-only cats sat encodings and verify the same decoded source. No next-token model or continuation probabilities are supplied here. Reserve sat cats/sat cat! for independent W4.

Keep partial failures and evidence limits

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

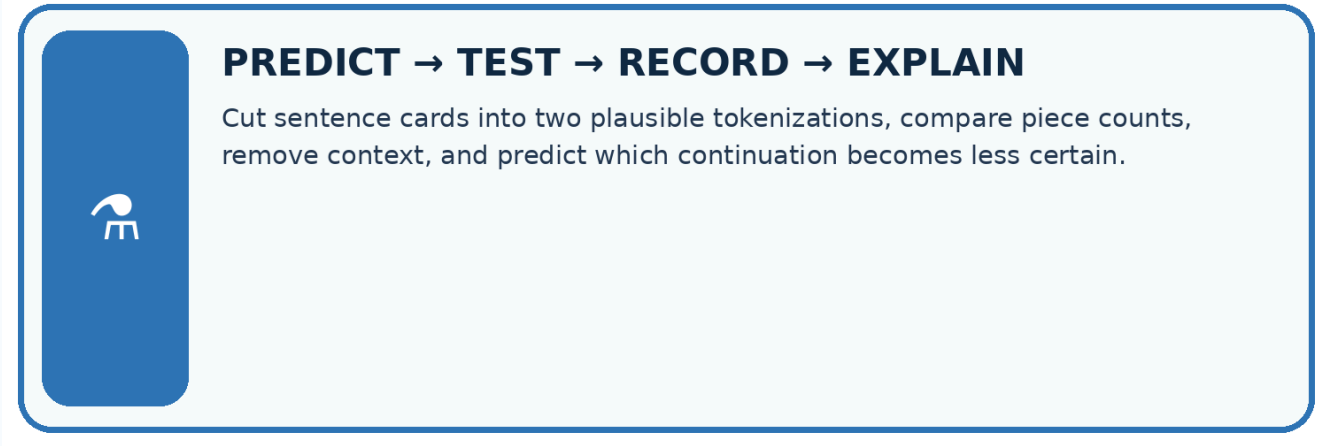
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the complete supplied toy encoder and four exact source strings. Record ordinary IDs, counting convention, decoded text or partial failure, then compare character-only vocabulary. No continuation model is supplied, so do not invent a probability change after removing **context**.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: trace the new phrase and its failing variant

> Count ordinary text tokens separately from an optional terminator.

The independent phrase pair

Under the original **vocabulary** encode sat cats, decode it and count text **token**s. Then trace sat cat! and state the exact failure position using zero-based character positions.

The complete original contract

Use the original case-sensitive vocabulary:1=cat,2=c,3=a,4=t,5=s,6=literal space,7=sat;0=end only if explicitly appended by the protocol. Greedily take the longest ordinary match. On failure retain partial IDs and report the zero-based source character position. Do not lowercase or remove punctuation.

Your first attempt and support record

> **My independent explanation, trace and limits**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Trace a new input independently

Week 25 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A classroom **token**izer splits sunflower into sun and flower. How many pieces result? Must every real **token**izer split it the same way?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Tokenize book descriptions before any text-based matching, while preserving the original source text.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Toy **token**izer A splits 'unhelpful' into un/help/ful; B keeps it whole. Count **token**s for each and explain why a three-word sentence need not use three **token**s.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Document the recommender’s text representation

> Keep original text separate from intentional transformations.

This week’s concrete contribution

For the fictional recommender preserve each original description and record any separately declared text preprocessing. Name the vocabulary version, matching rule, unknown-character policy and special-token convention before comparing counts or text features. Exact decoding does not establish factual correctness or good matching. This toy encoder is not documentation of a particular deployed language model.

Document the recommender’s text representation and unknown-**token** behavior rather than assuming every model counts words identically.

Keep the checkpoint, system and human choice

THIS WEEK’S CHECKPOINT **Token**ize book descriptions before any text-based matching, while preserving the original source text.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

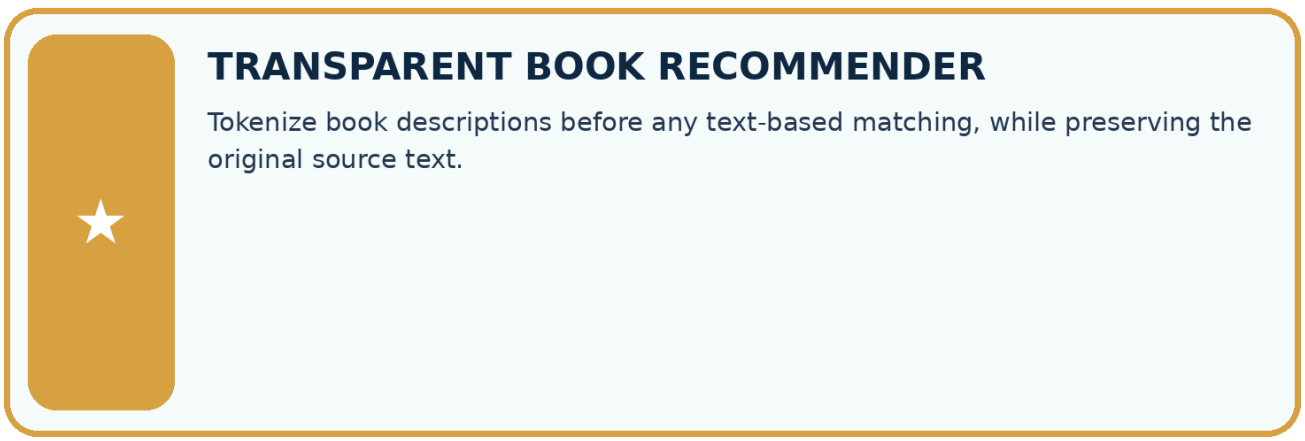
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: connect the count to its vocabulary and rule

> Explain what round-trip success does and does not establish.

Explain without notes

- Close your notes. Explain this goal in your own words: Encode and decode text under an explicit toy **vocabulary**

> **My deciding step, evidence and one remaining question**

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
token	A unit of encoded text processed by a language model, which may be a word, part of a word or punctuation.	
tokenization	Converting text into the token units of a particular tokenizer.	
context	Information available around a task that helps interpret or answer it.	
pattern	A relationship or regularity observed in examples.	

Terms for the connected explanation

Word	Meaning
Vocabulary	The permitted tokens and their identifiers
Encoding	Mapping source text into a token sequence under a specified rule

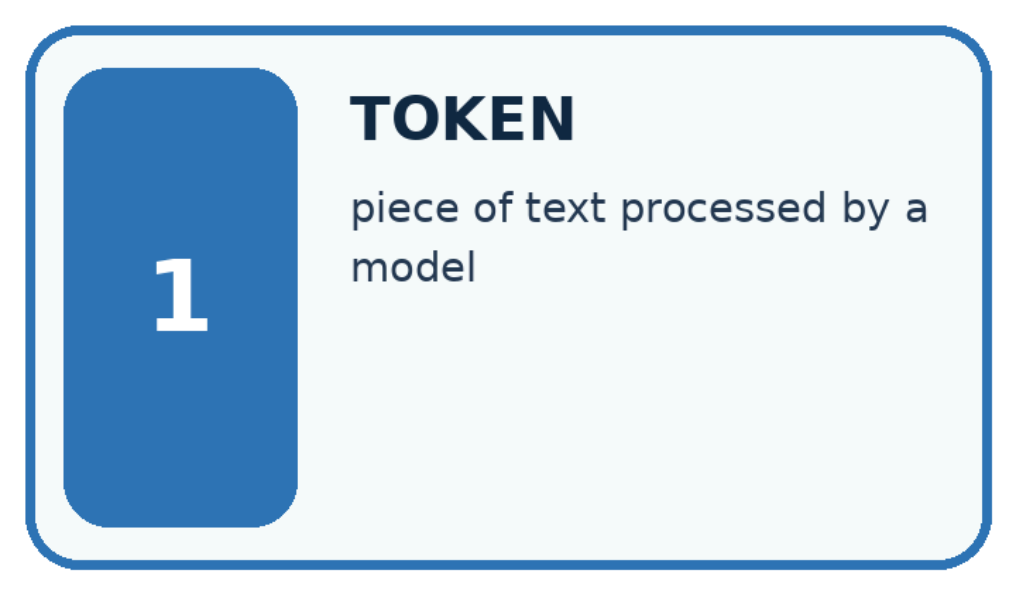
Word	Meaning
Decoding	Mapping a token sequence back to text under a specified rule

▼ **Earlier wording, alternatives and picture transcripts**

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

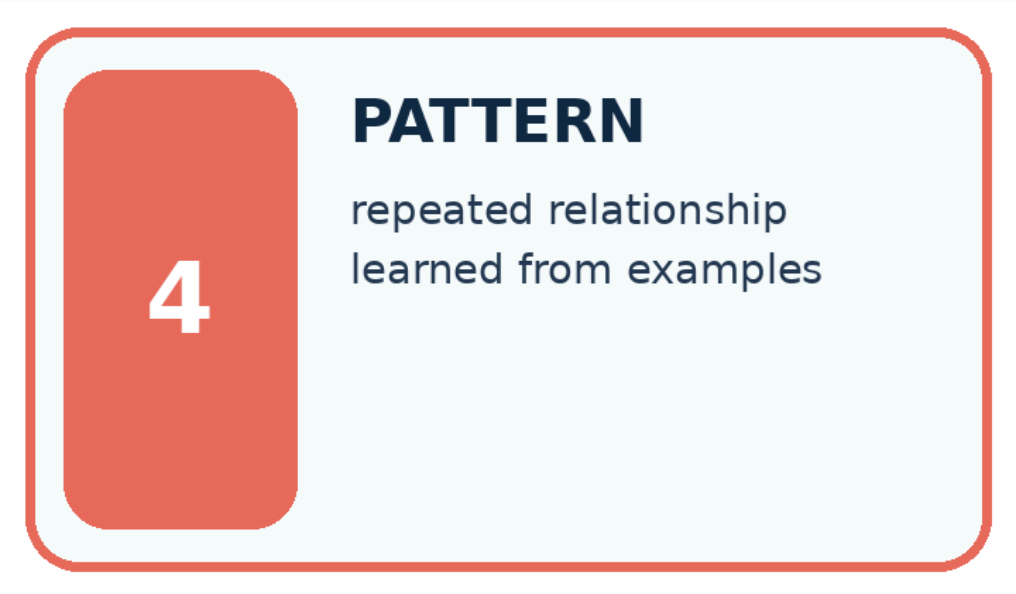
TOKENIZATION

rules for splitting text into pieces

3

CONTEXT

surrounding tokens available



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

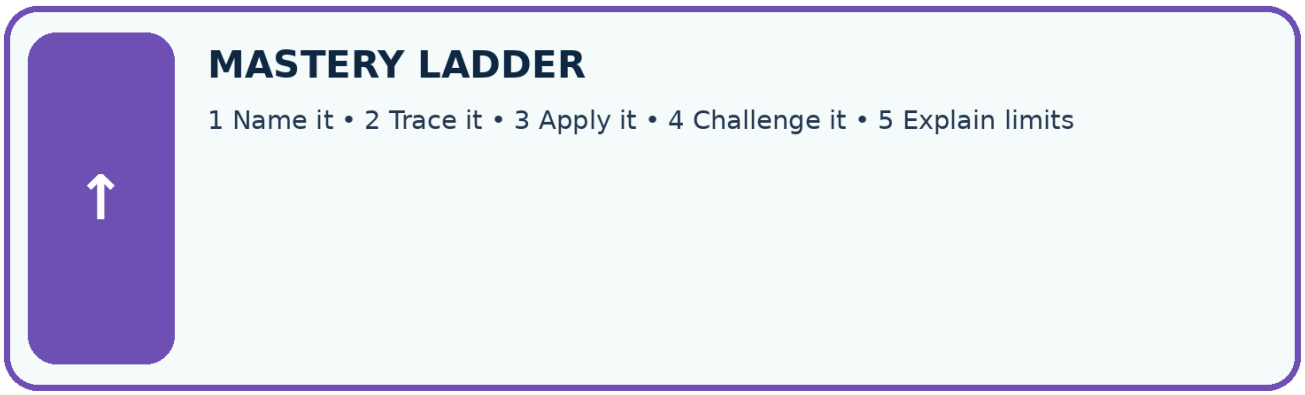
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Trace a new input independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

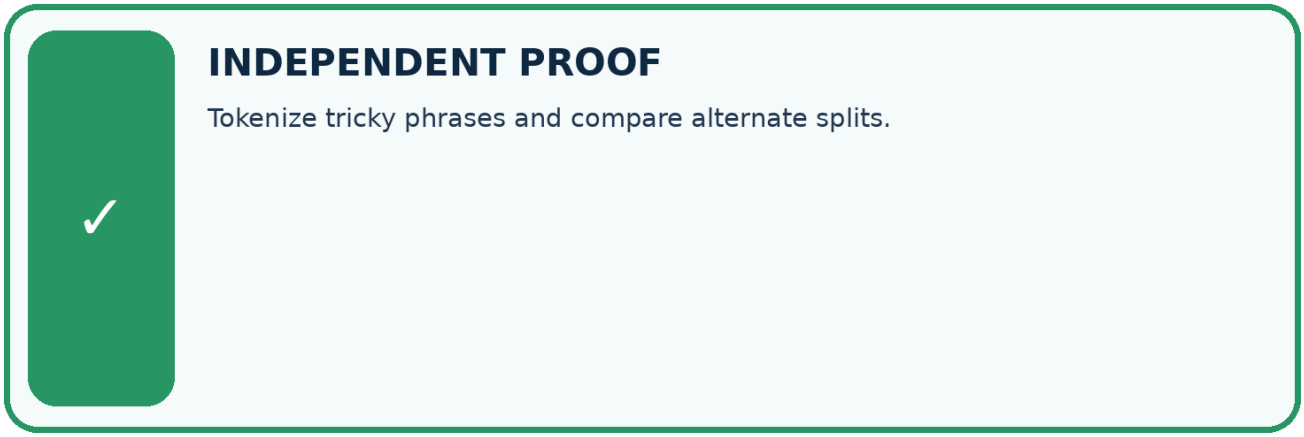
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Trace a new input independently is the selected independent case.



EXPLANATION CHECK Specify vocabulary, exact matching, ordinary versus special **token**s and explicit failure instead of silent repair.

> **Write or draw your reasoning and evidence.**

▼ **Extra practice: One word can have different token counts**

REFERENCE · SECTION 13

Extra practice: One word can have different token counts

> Try the question before opening its worked explanation.

Read the supplied case and question

Toy **token**izer A splits rainbows as [rain][bow][s]; B uses [rainbows].

Count **token**s and state the limit of the example.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Three versus one. **Token** count depends on **token**izer; **token**s need not equal words or letters.

Evidence or stage	Value or result
Tokenizer A	rain / bow / s
Tokenizer B	rainbows

▼ Earlier wording, alternatives and picture transcripts

Week 25 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

One word can have different token counts

Toy **token**izer A splits rainbows as [rain][bow][s]; B uses [rainbows].

Count **token**s and state the limit of the example.

Three versus one. **Token** count depends on **token**izer; **token**s need not equal words or letters.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Encode and decode text under an explicit toy vocabulary	Explain the deciding step, show a trace or test, and state one limit.
Explain how segmentation changes sequence length	Explain the deciding step, show a trace or test, and state one limit.
Handle unsupported characters without silently deleting information	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional **token**izer converts a short catalogue phrase into **token**s.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional **token**izer converts a short catalogue phrase into **token**s.
- Which labelled step matches this detail? For books book!, the ! at position 10 has no **token**: stop with error.
- 1. Unknown character
- 2. Decode
- 3. Encode phrase
- 4. Read prefix

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Unknown character For books book!, the ! at position 10 has no **token**: stop with error.

A **token** is defined by the **vocabulary** and **encoding** rule; it need not be one word or one character. Longest matching matters where **vocabulary** entries overlap. Counting text **token**s here does not count letters.

Preserve spaces during **decoding** so the reconstructed text matches the original exactly.

Changed-case question

- Changed-case practice: Use a character **vocabulary** covering b,o,k,s and space instead. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. The same phrase uses ten character **token**s.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The same phrase uses ten character **token**s. There are five letters, one space and four letters; its decoded text remains unchanged.

Repair and save your work

Review Feedback Loops (ai-2/24). Goal: Implement a deterministic recommender. Short practice: What extra field would reveal why comparing raw clicks is misleading?. Return to this week and explain what you changed.

Review Nearest Neighbor Reasoning (ai-2/22). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why do k=1 and k=3 disagree for the original query?. Return to this week and explain what you changed.

Review Sampling and Representation (ai-2/19). Goal: Define a question and target population. Short practice: Can **80%** response be reported as **80%** of the whole grade responding?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

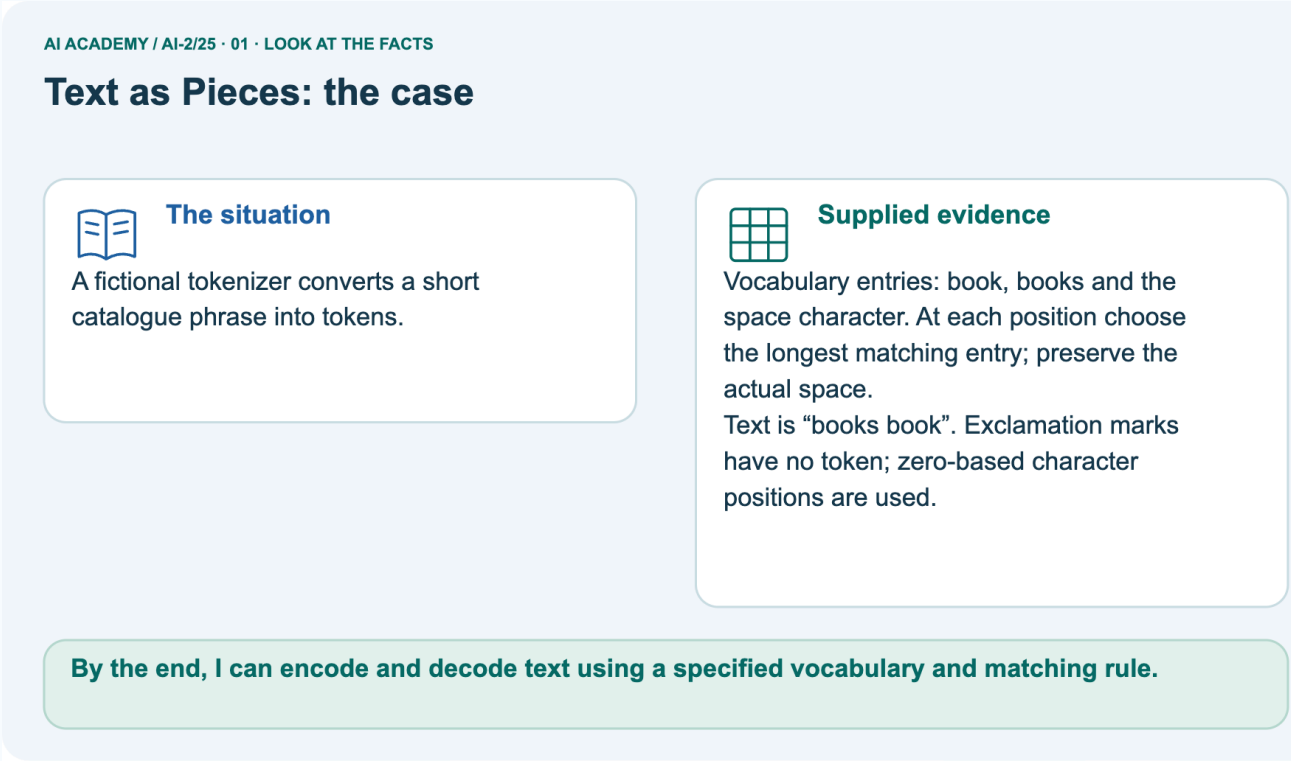
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Text as Pieces: the case



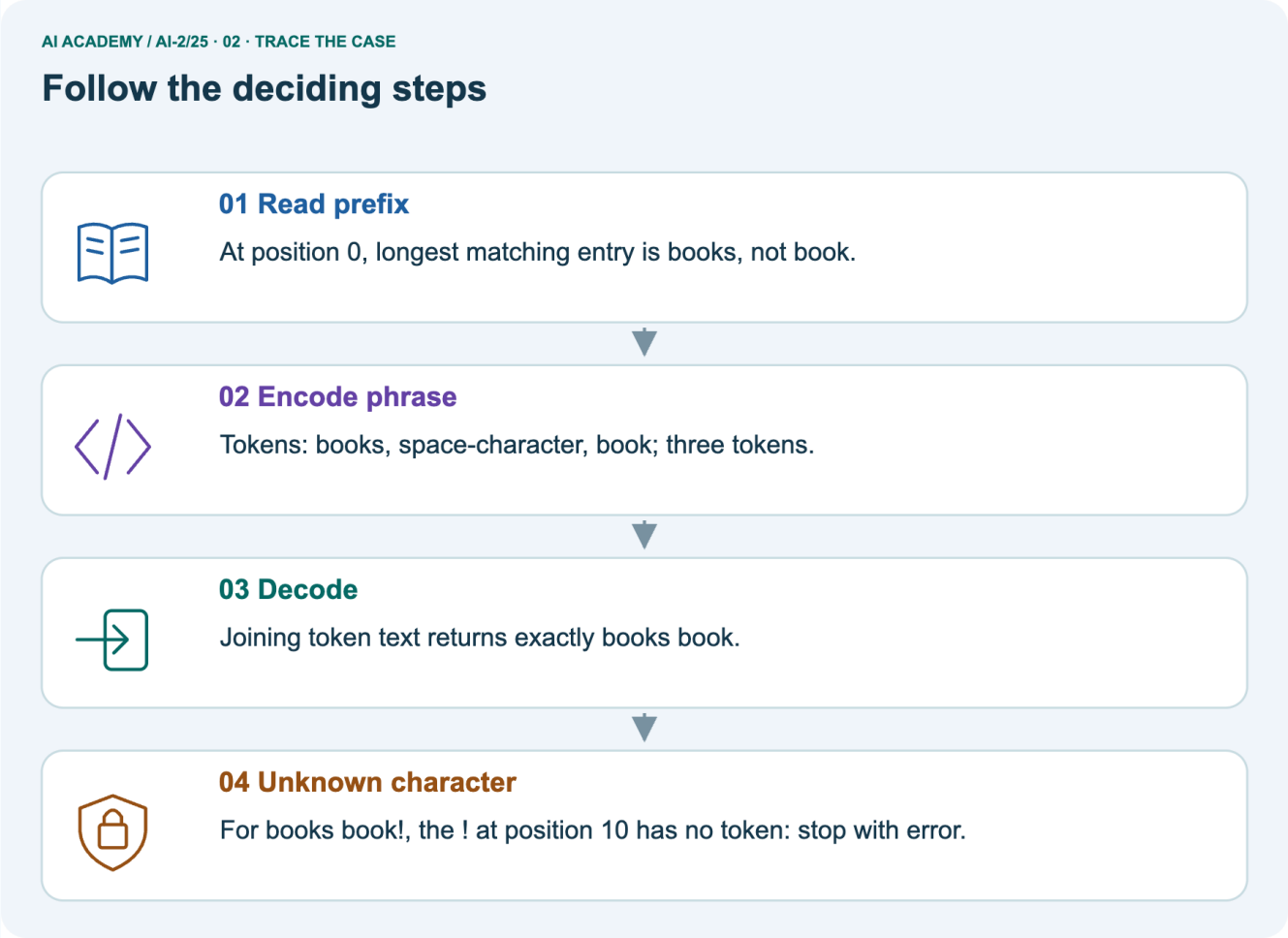
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional **token**izer converts a short catalogue phrase into **token**s. Supplied facts: **Vocabulary** entries: book, books and the space character. At each position choose the longest matching entry; preserve the actual space.

Text is “books book”. Exclamation marks have no **token**; zero-based character positions are used.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

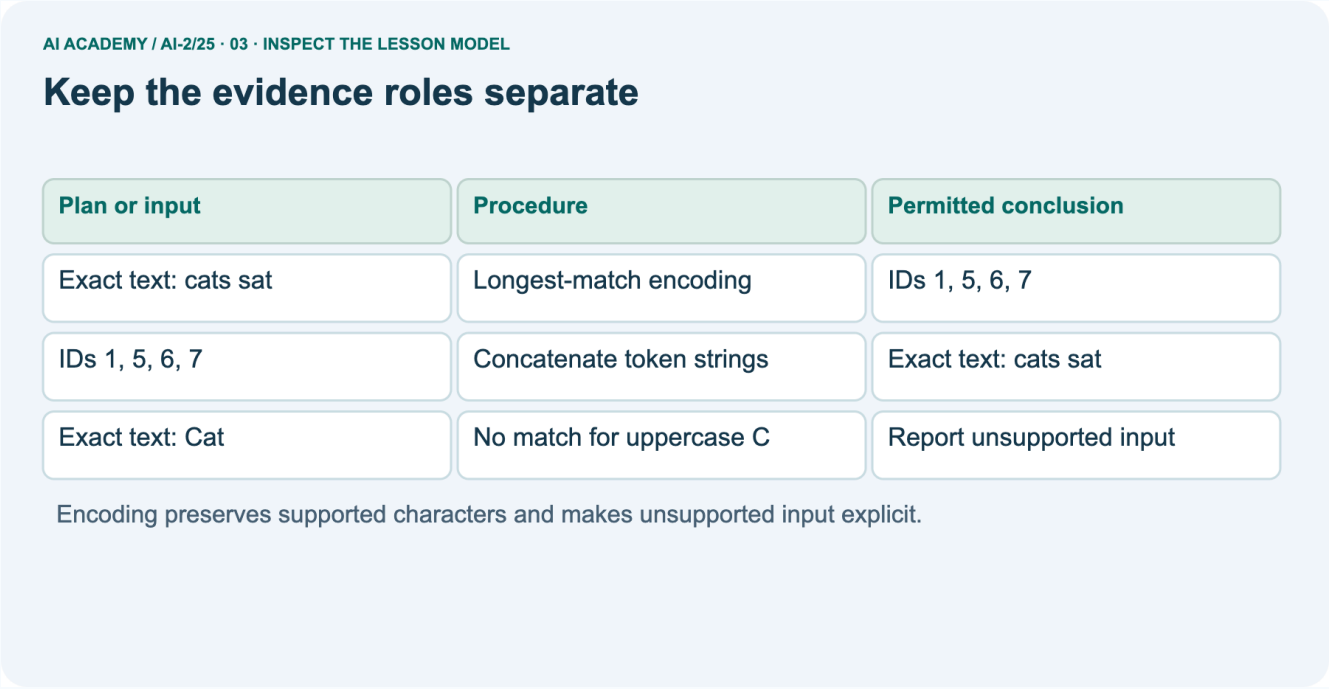


Follow the labelled picture with the matching explanation. Open the vector diagram

Read prefix: At position 0, longest matching entry is books, not book. → Encode phrase: **Token**s: books, space-character, book; three **token**s. → Decode: Joining **token** text returns exactly books book. → Unknown character: For books book!, the ! at position 10 has no **token**: stop with error.

Point to each numbered stage and explain which recorded **input** it uses.

03. Keep the evidence roles separate



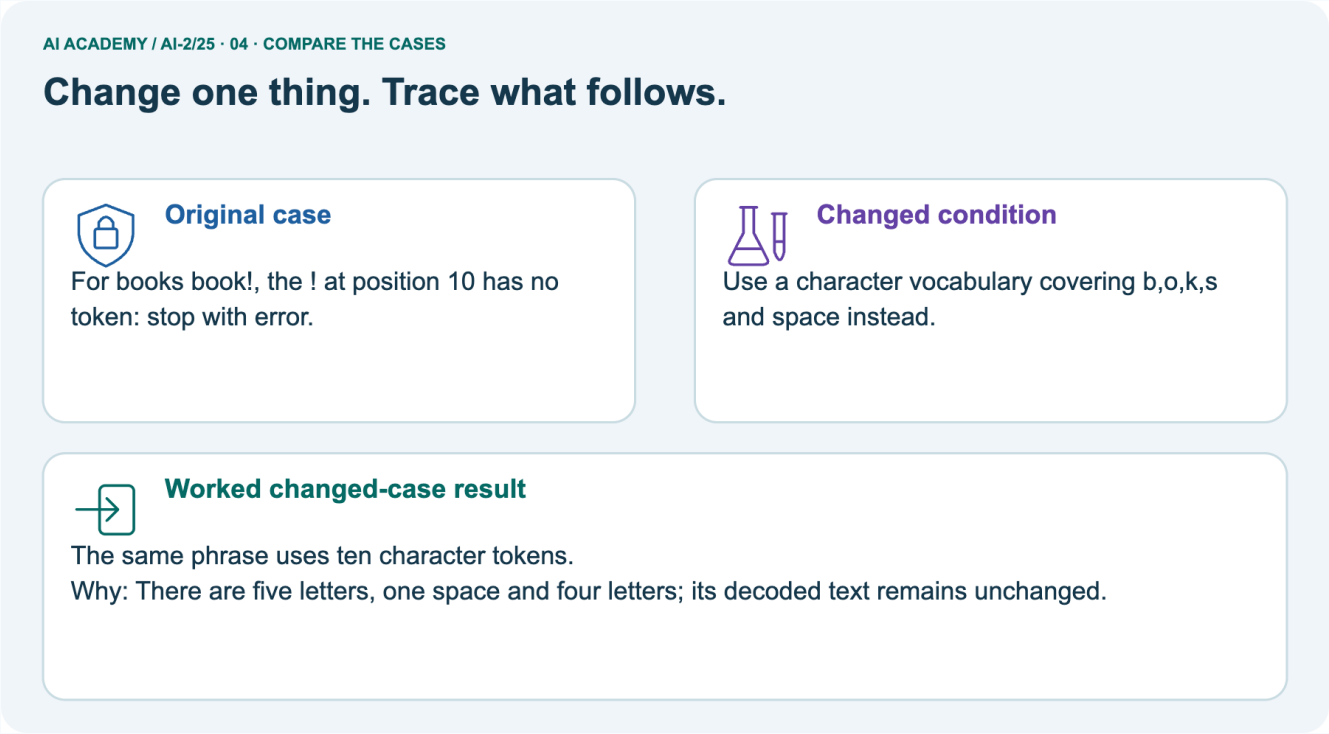
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. Exact text: cats sat | Longest-match **encoding** | IDs 1, 5, 6, 7. IDs 1, 5, 6, 7 | Concatenate **token** strings | Exact text: cats sat.

Exact text: Cat | No match for uppercase C | Report unsupported **input**

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



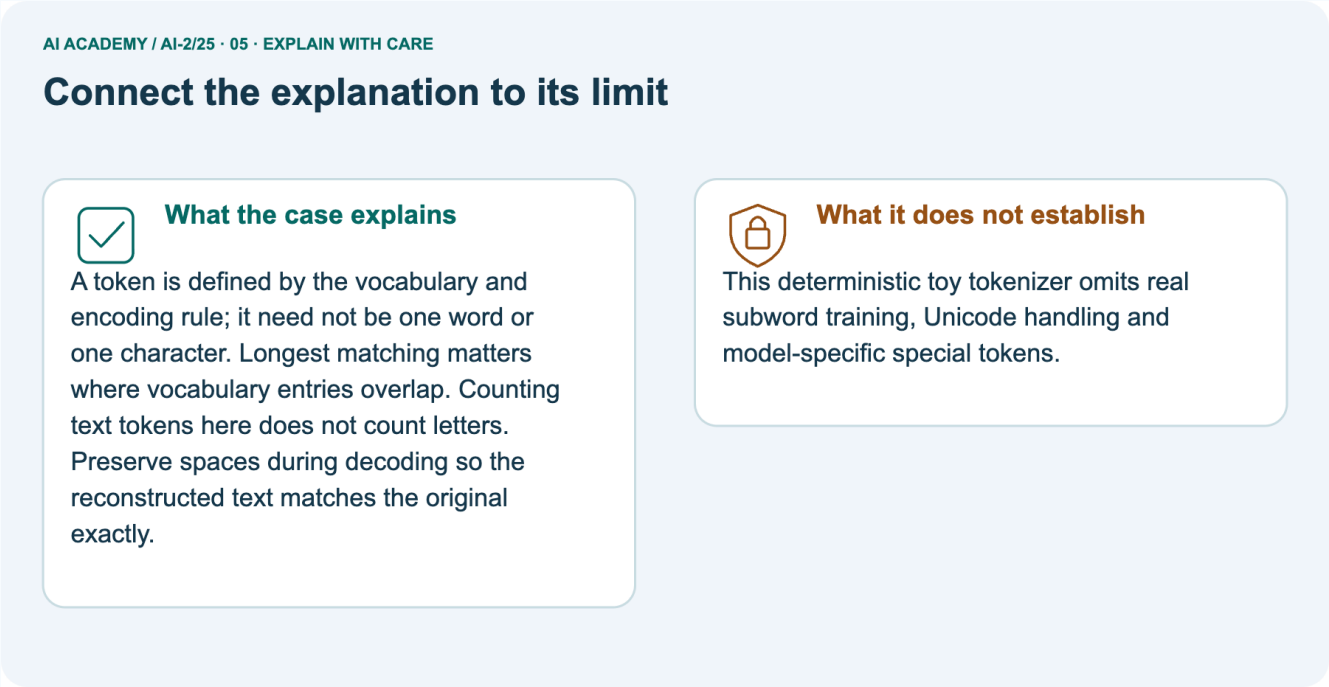
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: For books book!, the ! at position 10 has no **token**: stop with error. Changed condition: Use a character **vocabulary** covering b,o,k,s and space instead. Expected result: The same phrase uses ten character **token**s.

Reason: There are five letters, one space and four letters; its decoded text remains unchanged.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



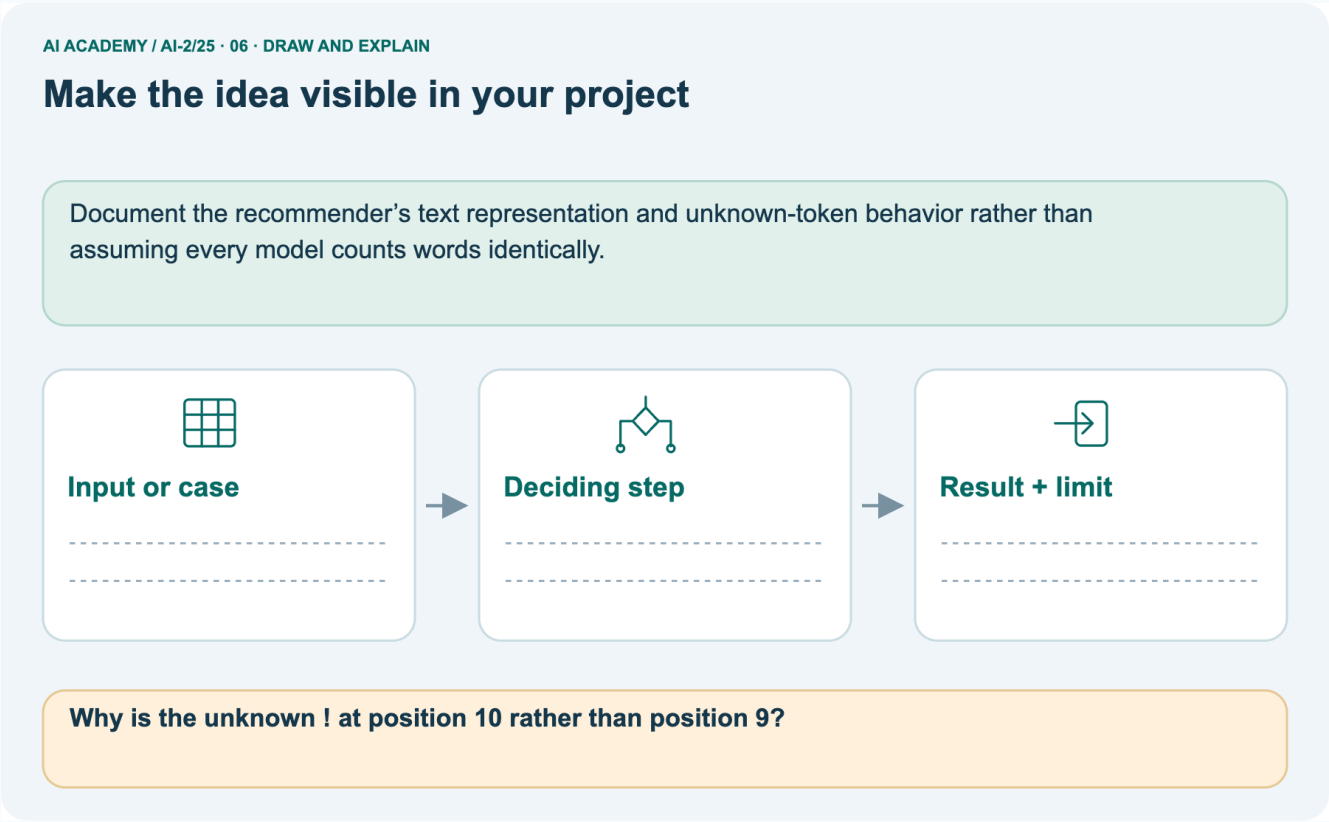
Follow the labelled picture with the matching explanation. Open the vector diagram

A **token** is defined by the **vocabulary** and **encoding** rule; it need not be one word or one character. Longest matching matters where **vocabulary** entries overlap. Counting text **token**s here does not count letters.

Preserve spaces during **decoding** so the reconstructed text matches the original exactly. Limit: This deterministic toy **token**izer omits real subword **training**, Unicode handling and model-specific special **token**s.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document the recommender's text representation and unknown-**token** behavior rather than assuming every model counts words identically. Draw your **input**, deciding step, result and limit.

Supported question: Why is the unknown ! at position 10 rather than position 9?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Text as Pieces: the case

02. Follow the deciding steps

03. Keep the evidence roles separate

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.